Project Description

1. **Our project**

   On busy streets in Downtown and Heliopolis delivery vans, ride hailing cars, buses, private cars all try to stop in the same small space beside the road in front of shops. Vans park in the middle of the street traffic gets stuck and nobody follows the traffic rules. The government also does not know how to solve and use the space in the right way .

   Smart curb is a mobile app that turns each space beside the road into a zone that drivers can book foe fixed time, it uses sensors and cameras that read car plates to make sure everyone follows the rules.

2. Goal

- Stop double parking and reduce traffic caused by vehicles fighting for space.
- Give the government real information about how the space is used.
- Make fines fair, and keep proof for every fine.
- Make deduction on anyone that park in any illegal place to decreases the traffic.

3. Who Will Use It

- **Traffic Planners:** draw the zones and choose their rules.
- **Delivery companies and drivers:** book places.
- **Ride-hailing and map apps (uber, didi, indrive):** get live information and pay for each stop.
- **Police:** need fines that are correct by law.
- **Shop owners:** can pay for free short stops for their customers.
- **Emergency services:** can empty a zone right away and cancel its reservations.

4. Main Features

- **Zones and rules:** Planners draw zones on a map and choose rules for each day and hour, for example deliveries from 7 to 11 am. They can copy rules from one street to another. The Head must approve changes, and they are announced 7 days early, except in emergencies.
- **Booking:** A driver reserves 10, 20 or 30 minutes from a phone in less than 30 seconds. The reservation is accepted only if the vehicle fits in the zone. If not, the system suggests another zone or time. Companies can reserve the same time every day, but no company can take more than 40% of a zone in busy hours.
- **Sensors and fines:** Sensors check if a space is used, and cameras read the plates in every zone. A vehicle with no reservation, or one that stays 5 minutes or more after its time, gets a violation.
- **Live data:** Map and ride-hailing apps get free live information about which zones are open and what the rules are.